From a molecular excitation to a collected photon

I think a good question to guide this intro is: if a molecule emits on our chip, how much of that light can reach the place where we want to use it? The path sounds straightforward. The molecule gives energy to a ring, the ring sends light into a bus waveguide, and a grating sends it off the chip toward a lens or fiber. But each step has its own physics, and making one step work well doesn't guarantee that the next one will.

A strong field at the molecule can help it put energy into the ring. That energy still has to leave through the useful bus output before it is lost elsewhere. At the grating, sending light upward is only part of the job -- the outgoing field also has to match the beam that the collection optics accept. I'd like to follow that whole path, keeping track of what each calculation actually tells us.

I'll use small analytical models so you can change one part of the system and see what happens. You can rotate the molecular dipole, change how long a photon stays in the ring, move the ring and bus farther apart, or change the light emitted along a grating. These models run locally, so there's no reason to spend cloud credits while getting comfortable with the ideas.

Tidy3D comes in when the geometry needs to tell us what the fields really are. It solves Maxwell's equations for the materials and shapes you give it. The quantum model answers a different set of questions: how the molecular excitation evolves, where the photon eventually goes, and what can be said about its statistics. The connection between those two descriptions is what I want you to get comfortable with.

$$|e,0\rangle \;\rightarrow\; \text{molecule and ring excitation} \;\rightarrow\; \text{bus photon} \;\rightarrow\; \text{collected mode}$$

The design target is 595 nm, with a TM-like grating coupler on X-cut lithium niobate. I haven't chosen a molecular species, host crystal, layer stack or collection optic for these examples. For now, the numbers are knobs you can turn. That gives you room to understand the tradeoffs before a particular material system fixes the available choices.

I'll start just after the molecule has been excited. There is one excitation in the molecule, no photon in the ring, and no laser still driving the system. That is what $|e,0\rangle$ means in the path above. From there, the energy can move into the ring, move back into the molecule, or leave through one of the loss channels. Later I'll come back to the laser pulse itself, because a pulse that is still present after emission can excite the molecule again. Experiments with molecules in cavities and near rings give useful examples of these processes. [R1, R2]

Before going further, I want to settle the factors of two. They come up often enough in cavity QED that it's good to set conventions quickly. I'll use κ for the rate at which the photon number, and therefore the stored optical energy, decays in the cavity. The field amplitude decays at half that rate, $\kappa/2$, because energy is proportional to amplitude squared.

For the molecule, γ_1 is the rate at which the excited-state population disappears. Losing the excitation also destroys optical coherence, but coherence can disappear without any population loss. That second process is pure dephasing, with rate γ_ϕ. By optical coherence, I mean the phase information in a superposition of the ground and excited states. Random shifts of the transition frequency can scramble that phase without changing the chance that the molecule is excited. Together they give the coherence decay rate $\gamma_2 = \gamma_1/2 + \gamma_\phi$. All these rates, and the angular frequencies below, have units of inverse time. Radians don't add a separate physical dimension.

$$\kappa = \kappa_i + \sum_j \kappa_j, \quad Q = \frac{\omega_c}{\kappa}, \quad \Delta\nu_c = \frac{\kappa}{2\pi}$$

$$\gamma_2 = \frac{\gamma_1}{2} + \gamma_\phi, \quad \Delta\nu_{\mathrm{hom}} = \frac{\gamma_2}{\pi}$$

Here κ_i is loss from the ring into channels such as absorption or unwanted radiation. The κ_j are the output channels I choose to keep track of separately, such as the two bus directions. The quality factor Q compares the optical oscillation frequency with the energy decay rate.

The linewidths in these equations are full widths at half maximum of an intensity spectrum. For the molecule, an exponentially decaying optical coherence produces the homogeneous linewidth shown above. This describes the width of one transition under those conditions. A collection of molecules with different transition frequencies, or one molecule whose frequency drifts over time, can produce a wider measured line for another reason. Some papers define κ or γ as amplitude decay rates instead, so I always check their definitions before borrowing a formula.

For a concrete scale, $g/2\pi = 100$ MHz means $g = 0.6283\,\mathrm{ns}^{-1}$. A 5 ns population lifetime means $\gamma_1 = 0.2\,\mathrm{ns}^{-1}$. With no pure dephasing, that molecule has a homogeneous optical linewidth of about 31.8 MHz. The optical frequency at 595 nm is much larger, about 504 THz. When calculating Q, use either angular frequency divided by angular-frequency linewidth or ordinary frequency divided by ordinary-frequency linewidth. Mixing them introduces a factor of 2π.

For classical fields, I'll write $\boldsymbol{E}(t) = \mathrm{Re}[\boldsymbol{E}_0 e^{-i\omega t}]$. The complex vector $\boldsymbol{E}_0$ keeps track of both the peak amplitude and the phase. This is what I mean by a phasor. With that convention, the average power crossing a surface is

$$P = \frac{1}{2}\mathrm{Re}\int_S (\boldsymbol{E}_0 \times \boldsymbol{H}_0^*) \cdot \hat{\boldsymbol{n}}\, dA\,.$$

If t is a transmission amplitude whose input and output modes are normalized to power, the transmitted power fraction is $|t|^2$. The same idea shows up in the grating model: if the amplitude decays at a certain rate along the guide, the power decays twice as fast. Keeping amplitude and power separate makes the later equations much easier to read.

A molecule doesn't emit because a tiny classical charge is literally moving back and forth inside it. The quantity that connects two molecular quantum states to the electric field is the transition dipole, $\boldsymbol{d}_{ge} = \left\langle g \left| \hat{\boldsymbol{d}} \right| e \right\rangle$. Its size tells us how strongly that transition interacts with light, and its direction tells us which field polarization it couples to.

There is also more than one way for an excited molecule to lose energy. Electronic states come with vibrational states. A selected zero-phonon line, or ZPL, is the optical transition without the additional vibrational excitation associated with a sideband. Other optical transitions can leave vibrational energy behind. The molecule may also decay without emitting a photon, or enter a long-lived state that temporarily stops the fluorescence. For the first model, I'll group the reference decay channels as follows.

$$\gamma_1^0 = \Gamma_{\mathrm{ZPL}}^0 + \Gamma_{\mathrm{PSB}}^0 + \gamma_{\mathrm{nr}}^0, \quad q = \frac{\Gamma_{\mathrm{rad}}^0}{\gamma_1^0}, \quad s = \frac{\Gamma_{\mathrm{ZPL}}^0}{\Gamma_{\mathrm{rad}}^0}$$

The superscript 0 means the reference environment before adding the cavity contribution. Quantum yield q is the fraction of population decay that produces any photon. The branching fraction s then asks what fraction of those radiative events use the selected ZPL. So the rate of the optical transition I want is $sq\gamma_1^0$.

You may see a Debye-Waller factor quoted for a molecule. Its precise meaning depends on how the spectrum was divided up, so I would check whether it gives the entire selected-line fraction needed here or only part of it. The useful number for the cavity model is the radiative rate of the actual transition that couples to the ring. Molecular experiments show why that distinction matters. [R1]

In a uniform, lossless dielectric with refractive index n_h, the macroscopic electric-dipole model relates that radiative rate to the dipole strength. It gives the following estimate. The microscopic field inside the molecular environment may differ from this smooth dielectric field, so a dipole inferred from data has to use a compatible host convention.

$$\Gamma^0_{\mathrm{ZPL}} = \frac{n_h \omega_e^3 |\boldsymbol{d}_{\mathrm{ZPL}}|^2}{3\pi\epsilon_0 \hbar c^3}, \quad |\boldsymbol{d}_{\mathrm{ZPL}}| = \left(\frac{3\pi\epsilon_0 \hbar c^3 s q \gamma_1^0}{n_h \omega_e^3} \right)^{1/2}.$$

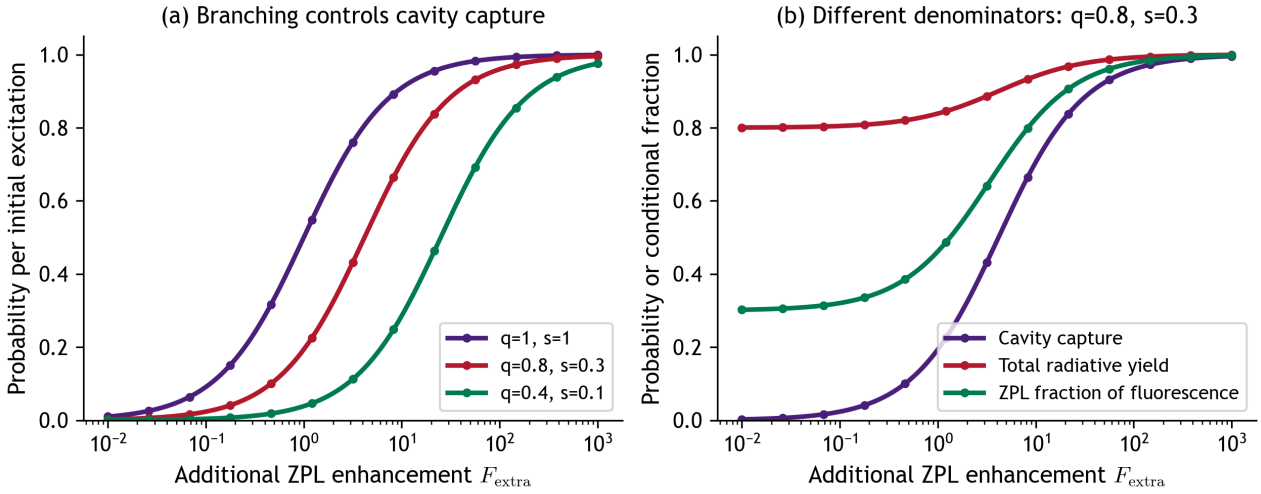
This is why a fluorescence lifetime by itself isn't enough to find the ZPL dipole. Two molecules can have the same total lifetime but send very different fractions of their emission into the transition of interest. I also need the quantum yield, the branching fraction, and a description of the host. Once g has been calculated using $\boldsymbol{d}_{\mathrm{ZPL}}$, the reduced strength of that line is already included in g^2. Multiplying the cavity rate by s again would apply the same reduction twice.

Here's a useful first picture of cavity enhancement. Suppose the cavity adds a decay channel on the ZPL, with rate $\Gamma_c = F_{\mathrm{extra}} \Gamma^0_{\mathrm{ZPL}}$, and leaves the other rates alone. Then the total population decay rate increases, and some fraction β_c of the excitations leave through that added channel:

$$\frac{\gamma_1^{\mathrm{new}}}{\gamma_1^0} = 1 + F_{\mathrm{extra}} s q, \quad \beta_c = \frac{F_{\mathrm{extra}} s q}{1 + F_{\mathrm{extra}} s q}.$$

The word extra matters here. F_{extra} counts only the rate added by the cavity. If someone instead defines F_{tot} as the entire ZPL rate divided by its reference value, the lifetime enhancement is $1 + sq(F_{\mathrm{tot}} - 1)$. The minus one removes the ZPL decay that was present before the cavity was added.

Physically, enhancing one weak branch can make a large difference to that branch while making a smaller difference to the total lifetime. For example, a molecule that rarely emits on the selected line still has all its other decay routes. The plot shows how the quantum yield and branching fraction affect that competition.



How an added ZPL channel changes the total decay rate and the fraction captured by the cavity. The quantum yield and selected-line fraction are adjustable inputs.

A classical dipole simulation is still very useful, even though the molecule is quantum mechanical. In this calculation, I define an oscillating current at the molecular position and ask what fields it produces. The source in these simulations is nothing more than that. It doesn't lose an excited-state population after emitting one photon, and it doesn't impose the photon statistics of a single emitter.

All we really care about right now though is the response of the electromagnetic environment to the molecule, and that can be modeled by a dipole in the weak-coupling electric-dipole limit. Coupling between this dipole model and its environment determines how readily a quantum transition can emit, as shown here. $\boldsymbol{p}_0$ is the classical dipole amplitude, and $\boldsymbol{G}$ is the Green tensor -- the field response to a point dipole, including its direction.

$$\boldsymbol{J}(\boldsymbol{r},t) = \dot{\boldsymbol{p}}(t)\delta(\boldsymbol{r}-\boldsymbol{r}_m), \quad \boldsymbol{E}_{\text{self}} = \mu_0\omega^2\boldsymbol{G}(\boldsymbol{r}_m,\boldsymbol{r}_m;\omega)\boldsymbol{p}_0$$

$$P_{\text{cl}} = \frac{\omega}{2}\text{Im}[\boldsymbol{p}_0^* \cdot \boldsymbol{E}_{\text{self}}] = \frac{\omega^3}{2\epsilon_0 c^2}\boldsymbol{p}_0^* \cdot \text{Im}\boldsymbol{G} \cdot \boldsymbol{p}_0$$

$$\Gamma_{\text{em}} = \frac{2\omega_e^2}{\hbar\epsilon_0 c^2}\boldsymbol{d}_{ge}^* \cdot \text{Im}\boldsymbol{G}(\boldsymbol{r}_m,\boldsymbol{r}_m;\omega_e) \cdot \boldsymbol{d}_{ge}\,.$$

The expression involving $\text{Im}\boldsymbol{G}$ asks how strongly the environment responds at the molecule's position, frequency and dipole direction. It is often described in terms of the local density of optical states. The physical question is how much electromagnetic response is available to this particular transition.

That response can include absorption as well as outgoing light. A nearby absorbing material can accept energy from the dipole without sending a useful photon toward the collection optics. So a flux monitor around the device and a calculation of total electromagnetic decay need not give the

same answer. I keep outgoing radiation, material absorption and the molecule's own nonradiative decay as separate channels.

The cleanest comparison is often a ratio. Run the same prescribed dipole in the device and in the chosen reference environment, using the same source amplitude and spectral normalization. Within the decay-rate approximation just described, the ratio of classical powers gives the ratio of electromagnetic decay rates:

$$\frac{\Gamma_{\mathrm{em}}}{\Gamma_{\mathrm{em,ref}}} = \frac{P_{\mathrm{cl}}}{P_{\mathrm{cl,ref}}}.$$

There is a factor of four hiding in an absolute comparison if the amplitude convention is left unspecified. With the peak phasor used here, a free-space classical dipole with amplitude $p_0 = d$ radiates $P_{\mathrm{cl}} = \hbar\omega\Gamma_0/4.$ Choosing $p_0 = 2d$ removes that factor. Both quantities are called a dipole, but the classical peak amplitude and the quantum transition matrix element enter different definitions. Using an identical source in the device and reference calculations makes that amplitude factor cancel.

In Tidy3D, PointDipole is this classical current source. You choose its position, polarization and time dependence. A broadband pulse is convenient because one run can probe the response over a range of frequencies. It doesn't mean the actual molecule emits that chosen pulse shape.

I would use the same source convention in a reference run, then separate the device output into the channels I care about. That tells me about the electromagnetic response. The quantum model still has to describe the initially excited molecule and the statistics of its emitted light. [R6, R7]

If the molecular line is narrow and the environment changes little across it, evaluating the response near the transition frequency can be enough. A broad molecular line samples a wider range of that response, so the spectral overlap matters. A narrow cavity can also give the excitation back to the molecule before the energy escapes. In that case, I keep the cavity as a dynamical degree of freedom and let cavity QED describe the exchange, rather than replacing everything by one irreversible decay rate.

To find g, I need the field associated with one quantum of excitation in the mode. The color scale on a mode plot is arbitrary until the mode has been normalized. If losses are small and the material's frequency dependence can be ignored in the stored-energy calculation, I can normalize the vector profile over the whole mode as below. The operator $\hat{a}$ removes a cavity photon, and the coefficient multiplying it sets the vacuum field scale.

$$\int \boldsymbol{u}^* \cdot \boldsymbol{\epsilon}_r \boldsymbol{u}\, dV = 1, \quad \hat{\boldsymbol{E}}^{(+)}(\boldsymbol{r}) = \sqrt{\frac{\hbar\omega_c}{2\epsilon_0}}\boldsymbol{u}(\boldsymbol{r})\hat{a}$$

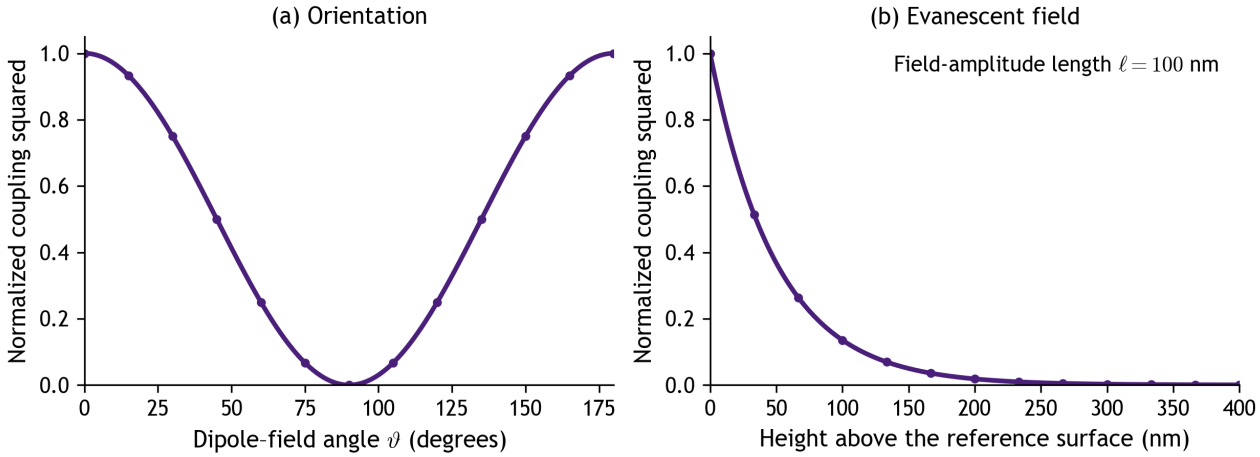
$$\boldsymbol{E}_{\text{vac}} = \sqrt{\frac{\hbar\omega_c}{2\epsilon_0}}\boldsymbol{u}, \quad g = -\frac{\boldsymbol{d}_{ge} \cdot \boldsymbol{E}^*_{\text{vac}}(\boldsymbol{r}_m)}{\hbar}.$$

The complex conjugate on the field comes from the emission term $g a^\dagger \sigma_-$ and the definition $\boldsymbol{d}_{ge} = \left\langle g \left| \hat{\boldsymbol{d}} \right| e \right\rangle$. I can change the overall phase assigned to one mode without changing the strength of its coupling, $|g|$. Relative phases become important when several modes or paths interfere.

The dot product is the part to picture first. A dipole couples to the component of the field along its own direction. In a locally linearly polarized field, rotating a real dipole through angle ϑ gives $g = g_{\max}\cos\vartheta$. A weak-coupling emission rate depends on $|g|^2$, so it follows $\cos^2\vartheta$. Moving the molecule away from an evanescent field gives another strong dependence. A useful toy is

$$g(z,\vartheta) = g(z_0,0)e^{-(z-z_0)/\ell_E}\cos\vartheta, \quad |g|^2 \propto e^{-2(z-z_0)/\ell_E}\cos^2\vartheta\,.$$

Here ℓ_E is an assumed decay length for the field amplitude. Moving up by one ℓ_E reduces the field, and therefore g, by a factor of e. A rate proportional to $|g|^2$ drops by e^2. This lets you see why height and alignment can matter so much. To put numbers on an actual site, I would use the vector field with the molecular host present, because the host can change both the field strength and its direction.



Dipole alignment and height in a locally linearly polarized field. The angle dependence comes from the vector projection. The exponential field decay is an assumed toy profile.

You can express the same field normalization using an effective mode volume. It compares the field throughout the mode with the field component seen by the molecule. For an isotropic host and a real, linearly polarized dipole direction $\hat{\boldsymbol{d}}$, the expression is

$$V_{\mathrm{eff}} = \frac{\int \boldsymbol{E}^* \cdot \boldsymbol{\epsilon}_r \boldsymbol{E}\, dV}{n_h^2 |\hat{\boldsymbol{d}} \cdot \boldsymbol{E}(\boldsymbol{r}_m)|^2}, \quad |g|^2 = \frac{|\boldsymbol{d}|^2 \omega_c}{2\hbar\epsilon_0 n_h^2 V_{\mathrm{eff}}}.$$

The integral in the numerator runs over the full mode. A small region with a very large field doesn't contain the whole ring photon's energy. In a uniform traveling-wave ring, every part of the circumference contributes. If the cross-section stays the same while the ring gets longer, the energy of one photon is spread over more length. The local field per photon then generally falls as the inverse square root of circumference.

The denominator also explains why a mode volume depends on the molecular site and direction. Put the molecule where its dipole sees almost no field, and the effective volume becomes large even if the field is strong somewhere else.

For a cavity that leaks strongly, or a material whose response varies appreciably with frequency, normalizing the mode takes more care. The stored energy depends on material dispersion, and a leaking resonance isn't a closed, energy-conserving normal mode. Quasinormal-mode methods handle those open resonances and their complex frequencies. I'll use the elementary normalization for the toy here. Before applying it to a resonance extracted from FDTD, I would check whether the loss and dispersion assumptions are adequate. [R8]

A ring resonates when the phase accumulated in one trip around it is an integer multiple of 2π. The wave then returns with the phase needed to reinforce itself. For circumference $L = 2\pi R$, that condition is $\beta L = 2\pi m$. The spacing between nearby resonances depends on how the phase changes with frequency, which is described by the group index n_g:

$$\beta(\omega_m)L = 2\pi m, \quad t_{\mathrm{rt}} = \frac{n_g L}{c}, \quad \mathrm{FSR}_\nu \simeq \frac{c}{n_g L}.$$

The ring and bus exchange light because their guided fields extend beyond the material boundaries. When the guides are close enough, these evanescent fields overlap. A model of two locally parallel, lossless guides already captures the main idea. Let their amplitudes be normalized so that $|a_r|^2$ and $|a_b|^2$ are powers. Their evolution along the coupling region is

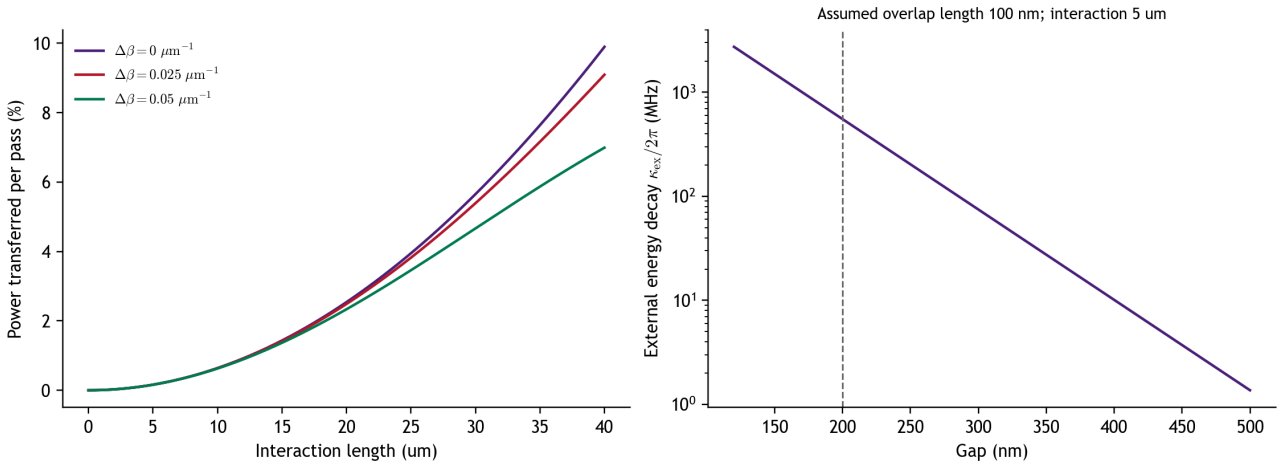
$$\frac{d}{d\zeta}\begin{pmatrix} a_r \\ a_b \end{pmatrix} = i\begin{pmatrix} \delta\beta/2 & \mu \\ \mu^* & -\delta\beta/2 \end{pmatrix}\begin{pmatrix} a_r \\ a_b \end{pmatrix}.$$

The coordinate ζ is distance along that region. The coefficient μ tells us how strongly the guides exchange light per unit distance. The mismatch $\delta\beta$ tells us how quickly their phases slip relative to each other. Both have inverse-length units. Start with light in the ring guide, and the amplitudes can exchange power as they propagate together. If the coefficients are constant over an interaction length L_{int}, the transferred power fraction is

$$K = \frac{|\mu|^2}{\Omega_\zeta^2}\sin^2(\Omega_\zeta L_{\mathrm{int}}), \quad \Omega_\zeta = \sqrt{|\mu|^2 + (\delta\beta/2)^2}.$$

When the propagation constants match, the contributions to the transferred field keep building in phase. This lossless model can then transfer all the power at the right interaction length. With a mismatch, power transferred at different positions doesn't keep adding constructively. Increasing the length alone can no longer give complete transfer.

For the gap dependence, I'll use $\mu(d) = \mu(d_0)e^{-(d-d_0)/\ell_{\mathrm{ov}}}$. It represents the overlap becoming smaller as the guides move apart, with an assumed length scale ℓ_{ov}. The bus-coupler toy lets you vary this gap law, the mismatch and the interaction length. An electromagnetic model would supply those quantities for the curved ring and its actual cross-section.



Two-guide coupling as a function of interaction length and gap. A propagation mismatch limits power transfer. The gap dependence uses an assumed exponential overlap, and the transferred fraction per pass is converted to a ring decay rate.

Now suppose a fraction K of the ring power leaves through the bus every round trip. If the intrinsic power attenuation along the ring is α_P, I can convert these spatial losses into the time-dependent rates used in cavity QED:

$$\kappa_{\mathrm{ex}} = -\frac{\ln(1-K)}{t_{\mathrm{rt}}}, \quad \kappa_i = \alpha_P v_g, \quad v_g = \frac{c}{n_g}.$$

For small K, the logarithm gives $\kappa_{\mathrm{ex}} \simeq K/t_{\mathrm{rt}}$. That is a useful way to read the result: the loss rate is approximately the fraction lost per pass divided by the time between passes. The same coupler therefore gives a smaller escape rate on a longer ring, because a photon encounters it less often. K is a dimensionless fraction, while κ_{ex} has units of inverse time.

Using one longitudinal ring resonance is reasonable when the relevant linewidths and couplings are small enough compared with the free spectral range, or when another physical reason removes the nearby modes from the problem. If several resonances overlap the molecular spectrum, they can all participate. I would then keep those modes explicitly or describe the environment with its frequency-dependent response.

There is one geometrical point to keep in mind for X-cut LN. The crystal axes are fixed in the lab, while the direction of propagation changes as light travels around the ring. Rotating the material tensor along with the ring tangent would describe a different crystal. A TM-like field may stay mostly vertical, but its smaller components and any polarization mixing still depend on the full vector solution. Those are the fields I would eventually use to calculate the local molecular coupling.

I'll first keep one molecular transition and one ring mode. The operator σ_- lowers the molecule from its excited state to its ground state, and $a^\dagger$ adds a photon to the ring. The reverse operators undo those changes. It is convenient to describe the system in a frame rotating at the molecular frequency, so the optical carrier doesn't appear in every equation. With $\Delta = \omega_c - \omega_e$, the Hamiltonian is

$$\frac{H}{\hbar} = \Delta a^\dagger a + g a^\dagger \sigma_- + g^* a \sigma_+ .$$

The two terms containing g exchange one molecular excitation for one ring photon, and back again. They keep the excitation inside the molecule–ring system. The other processes let energy escape into channels that I don't follow coherently, such as the bus output or background radiation.

I'll describe those channels with a Lindblad master equation. The density matrix ρ contains the populations and their quantum coherences. For an operator L, the combination $\mathscr{D}[L]\rho = L\rho L^\dagger - \{L^\dagger L, \rho\}/2$ accounts for the corresponding jump and its effect on the remaining state. Here the braces denote an anticommutator. The equation is

$$\dot{\rho} = -i[H/\hbar, \rho] + \kappa_i \mathscr{D}[a]\rho + \sum_j \kappa_j \mathscr{D}[a]\rho + \gamma_1 \mathscr{D}[\sigma_-]\rho + \frac{\gamma_\phi}{2}\mathscr{D}[\sigma_z]\rho .$$

The factor $\gamma_\phi/2$ is chosen so that pure dephasing damps the molecular optical coherence at rate γ_ϕ. In the one-excitation calculation, $2\gamma_\phi \mathscr{D}[|e\rangle\langle e|]$ gives the same dephasing. The population rate γ_1 includes decay outside the cavity mode that I have kept explicitly. If a classical calculation includes the cavity peak in a total decay rate, I have to separate it before using that rate as γ_1. Otherwise the same cavity contribution enters through both γ_1 and the g, κ dynamics.

This is a Markov model: the loss channels are represented by fixed rates, with no memory of an excitation that has already left. A coherent reflection that comes back later would need a different treatment.

Starting from $|e,0\rangle$ with no further drive, only two states can still hold the excitation: $|e,0\rangle$ and $|g,1\rangle$. Once it leaves, the molecule and cavity are both in their ground state. For this one-mode problem, I can choose the basis phase so that g is real. Let $p_e = \rho_{e0,e0}$ be the excited-molecule probability, $p_c = \rho_{g1,g1}$ the cavity-photon probability, and $c = \rho_{e0,g1}$ the coherence between them. Their equations are

$$\dot{p}_e = -\gamma_1 p_e - 2g\,\mathrm{Im}c, \quad \dot{p}_c = -\kappa p_c + 2g\,\mathrm{Im}c,$$

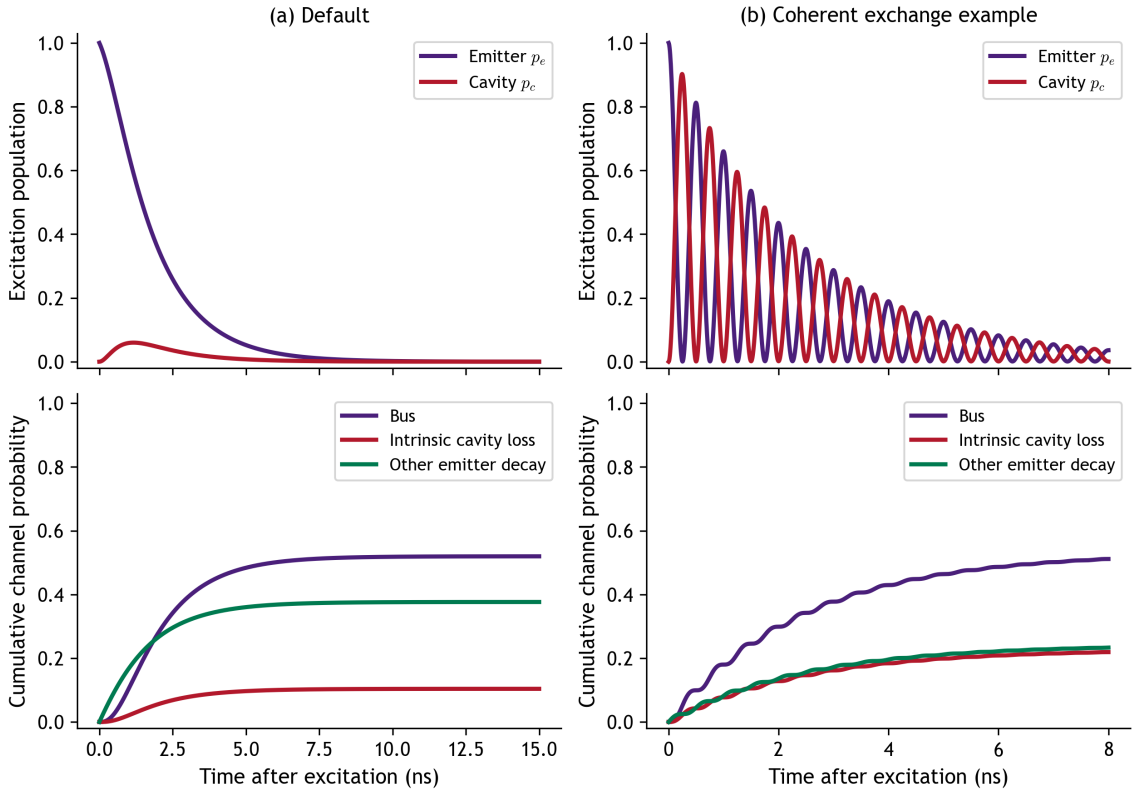
$$\dot{c} = -(A - i\Delta)c + ig(p_e - p_c), \quad A = \frac{\kappa + \gamma_1}{2} + \gamma_\phi .$$

The same exchange term appears with opposite signs in the two population equations. What leaves the molecule enters the cavity, and vice versa. The imaginary part of c controls that flow. This is the piece that a one-way decay picture misses when a cavity photon can be reabsorbed.

The sign in the detuning term follows $\Delta = \omega_c - \omega_e$ and the stated definition of c. Using the opposite convention would reverse that sign, while leaving the eventual probabilities below unchanged. To follow where the excitation ends up, I also keep cumulative probabilities for the loss channels:

$$\dot{P}_i = \kappa_i p_c, \quad \dot{P}_j = \kappa_j p_c, \quad \dot{P}_{\mathrm{bg}} = \gamma_1 p_e,$$
$$p_e + p_c + P_i + \textstyle\sum_j P_j + P_{\mathrm{bg}} = 1 .$$

Each output probability grows at its jump rate. For example, $\kappa_j p_c(t)\, dt$ is the probability that the photon leaves through output j during a short interval dt. Integrating that rate gives the probability of using that output at any time. The conservation equation says that the excitation is either still in the molecule, still in the cavity, or has left through one of the listed channels. Pure dephasing changes the coherence but doesn't remove the excitation, so it doesn't get a separate lost-photon probability. These are probabilities following one prepared excitation, a steady-state laser-probe experiment asks a different question. The impulse-excited cavity-QED literature discusses the same competition between exchange, reabsorption and escape. [R3]



Time evolution from one excited molecule and an empty cavity. The plots show the excitation moving between molecule and cavity, together with the probability accumulated in each output. The weak and strong exchange cases use the parameters saved with the data.

For collection, I often want one number: what is the chance that the excitation eventually leaves through the useful output? The full time trace is interesting, but it isn't necessary to find that number in this model. I can integrate the equations over the entire emission event.

Write $I_e = \int_0^\infty p_e\,dt$, $I_c = \int_0^\infty p_c\,dt$ and $I_{ec} = \int_0^\infty c\,dt$. The first two are the total time spent in the molecular and cavity states, weighted by their probabilities. Assuming the excitation eventually leaves, the initial and final coherences are both zero. Integrating the coherence equation then gives

$$I_{ec} = \frac{ig(I_e - I_c)}{A - i\Delta}, \quad 2g\,\mathrm{Im}\,I_{ec} = \mathscr{R}(I_e - I_c), \quad \mathscr{R} = \frac{2g^2 A}{A^2 + \Delta^2}.$$

The quantity $\mathscr{R}$ has rate units and contains the coupling, detuning and coherence decay. It is a convenient way to write the integrated result. The population equations give two more relations:

$$\kappa I_c = \mathscr{R}(I_e - I_c), \quad 1 = \gamma_1 I_e + \kappa I_c.$$

The first relation balances the total cavity escape with the net transfer from the molecule. The second says that the one initial excitation eventually leaves through molecular background decay or a cavity channel. Solving those two equations gives

$$P_{\mathrm{cav}} = \frac{\kappa \mathscr{R}}{\kappa \gamma_1 + (\kappa + \gamma_1)\mathscr{R}}, \quad P_j = \frac{\kappa_j}{\kappa} P_{\mathrm{cav}}.$$

Here P_{cav} includes every loss route represented by κ, including intrinsic cavity loss. The useful output j gets its share κ_j/κ.

Although $\mathscr{R}$ looks like a transfer rate, I haven't assumed that the populations follow rate equations at each instant. The time-dependent model still includes coherent exchange and reabsorption. Those dynamics have simply been integrated out of the final probability. Replacing the coherence by its instantaneous steady value, $\dot{c} = 0$, would need an additional argument that it relaxes much faster than the populations.

On resonance, with no pure dephasing, the answer becomes particularly easy to read. Define the cooperativity $C_0 = 4g^2/(\kappa\gamma_1)$. It compares coherent coupling squared with the product of the two population-loss rates. Then

$$P_{\mathrm{cav}} = \frac{\kappa}{\kappa + \gamma_1} \frac{C_0}{1 + C_0}.$$

If $\kappa \gg \gamma_1$, the first factor is nearly one, leaving the familiar $C_0/(1 + C_0)$. The extra factor matters when the cavity holds the photon for a long time. During that time, the energy can return to the molecule, where background decay gets another opportunity to take it away. Making the cavity longer lived therefore doesn't guarantee that more excitations reach its output.

There are also useful checks at the edges of the formula. With $g = 0$, the molecule cannot put a photon into the cavity. With $\kappa_j = 0$, no photon can leave through output j.

The quantum toy calculates the density-matrix evolution independently and compares the integrated outputs with this formula. You can use the time traces to see how two cases with similar final probabilities can get there through quite different dynamics.

The Purcell-rate picture comes from a limit in which the cavity responds much faster than the molecule–cavity exchange and the molecular coherence. The cavity field then follows the molecular amplitude closely enough that I can remove it as a separate dynamical variable. This is often called adiabatic elimination. The molecule acquires an extra population decay rate

$$\Gamma_c(\Delta) \simeq \frac{g^2\kappa}{\Delta^2 + (\kappa/2)^2}, \quad \Gamma_c(0) \simeq \frac{4g^2}{\kappa}.$$

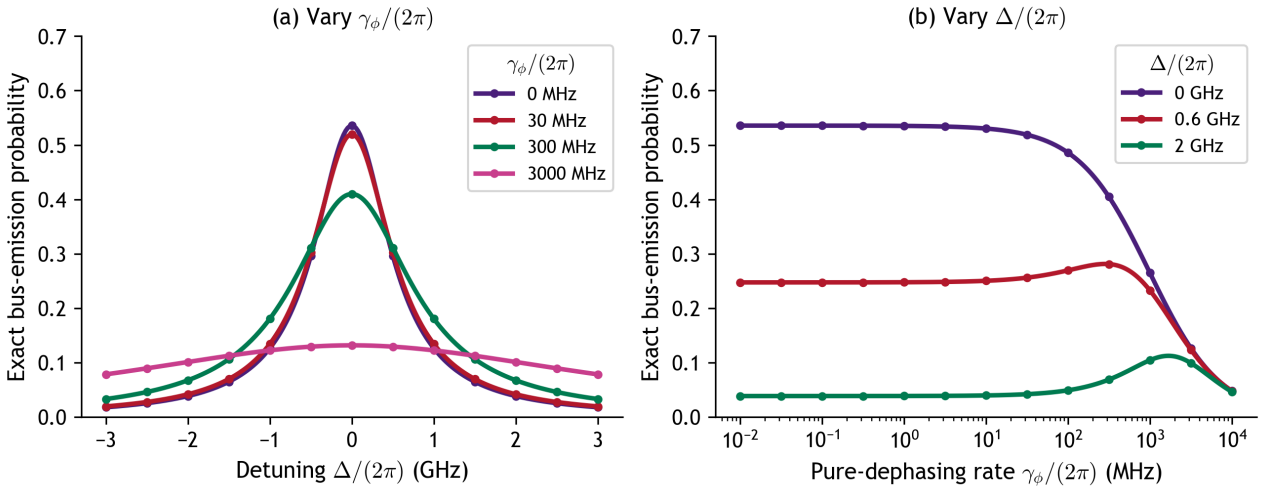
On resonance, the rate grows as g^2 and falls as the cavity linewidth κ increases. Combining g with the vacuum-field normalization and comparing with the homogeneous reference transition gives the usual single-mode Purcell expression. At a matched field maximum and dipole orientation,

$$F_{\mathrm{extra}} \simeq \frac{3}{4\pi^2}\left(\frac{\lambda_0}{n_h}\right)^3 \frac{Q}{V_{\mathrm{eff}}}.$$

The ratio Q/V_{eff} captures two useful ideas: keeping the light around longer and putting more field at the molecule. But the rate formula assumes an isolated resonance, weak enough coupling for the cavity to follow the molecule, and a molecular line narrow enough to sample the cavity response this way. Position, orientation and detuning still enter through the local field and the spectral overlap. Before interpreting a quoted Q/V as a lifetime change, I would check those assumptions and the molecular branching discussed earlier.

If the molecular line is broad, a narrower cavity sees only part of it. In the integrated model, the relevant width is $A = \kappa/2 + \gamma_2$. Once the molecular width dominates, raising Q no longer gives the improvement suggested by a narrow-emitter formula.

Dephasing has an interesting effect away from resonance. It can broaden the molecular response enough to increase overlap with a detuned cavity, even while reducing optical coherence. So more dephasing can increase a particular output probability in one range of parameters and decrease it in another. The detuning and dephasing plots are useful for seeing that directly.



How detuning and pure dephasing change the eventual output probability in the one-excitation model. All curves use the same definitions of molecular population decay and cavity energy decay.

To see when a decay-rate description starts to miss the exchange dynamics, look at the two complex frequencies of the molecule–cavity amplitude equations. With no pure dephasing, they are

$$\tilde{\omega}_{\pm} = \frac{\omega_c + \omega_e}{2} - i\frac{\kappa + \gamma_1}{4} \pm \sqrt{g^2 + \left[\frac{\Delta}{2} - i\frac{\kappa - \gamma_1}{4}\right]^2}.$$

The real parts of $\tilde{\omega}_{\pm}$ give the two mode frequencies. Their imaginary parts describe amplitude decay. At zero detuning, the square root develops a real splitting when $g > |\kappa - \gamma_1|/4$. Whether two peaks can actually be resolved also depends on their widths and on which output is measured.

It is useful to keep three questions separate: is the cooperativity large, does the excitation visibly oscillate between molecule and cavity, and does the measured spectrum resolve two peaks? They are related, but they don't have the same threshold. With pure dephasing, I use the full density-matrix evolution. A decaying amplitude by itself misses the way dephasing changes coherence while leaving population behind. The linear operator that evolves the density matrix is usually called the Liouvillian.

There is another question beyond getting a photon out: will photons from repeated events interfere well with one another? That depends on how reproducible and coherent their emitted quantum states are. For an unfiltered photon with exponential population decay and Markov pure dephasing, an approximate indistinguishability is

$$\mathcal{I} \simeq \frac{\Gamma_{\mathrm{pop}}}{\Gamma_{\mathrm{pop}} + 2\gamma_\phi}.$$

Here Γ_{pop} is the relevant population decay rate in that limit. Faster emission can help because there is less time for dephasing before the photon leaves. Cavity filtering, strong exchange, vibrational sidebands and slow frequency drift can change this picture.

To calculate the photon coherence more generally, I need a two-time field correlation: how the emitted field at one time is related to the field at another. Within the Markov model, the quantum regression theorem evolves that correlation using the same density-matrix dynamics. An FDTD flux spectrum gives useful information about the passive structure, but it doesn't supply this quantum correlation on its own. [R14]

So far I've drawn the ring as one mode. A ring usually has two nearly degenerate traveling directions, clockwise and counterclockwise. A one-mode description can represent one selected physical mode or a particular combination of the two, but I have to say which. To keep both directions explicitly, the Hamiltonian becomes

$$\frac{H}{\hbar} = \sum_{\mu=\pm} \Delta_\mu a_\mu^\dagger a_\mu + \sum_{\mu=\pm} (g_\mu a_\mu^\dagger \sigma_- + g_\mu^* a_\mu \sigma_+) + J a_+^\dagger a_- + J^* a_-^\dagger a_+ .$$

The coefficient J describes coherent backscattering between the directions. If the uncoupled frequencies are equal, that mixing gives two standing-wave modes separated by $2|J|$. A molecule can be near a field maximum of one standing wave and a node of the other. So if a ring spectrum shows a doublet, I would inspect the two fields before assigning a molecular coupling to each peak.

$$g_B = \sqrt{|g_+|^2 + |g_-|^2}, \quad a_B^\dagger = \frac{g_+ a_+^\dagger + g_- a_-^\dagger}{g_B} .$$

When the two directions are degenerate, have identical cavity dynamics and don't mix, the molecule couples only to the combination $a_B^\dagger$ written above. This is the bright mode. Its coupling is g_B. The orthogonal combination is dark to that molecular transition -- the two field contributions cancel in the coupling.

Backscattering, unequal frequencies or unequal losses can couple the bright and dark combinations. In the toy comparisons, I hold g_B fixed when switching between one mode and two directions. That way, a change in collection comes from the output geometry or dynamics, rather than from quietly increasing the total coupling by $\sqrt{2}$.

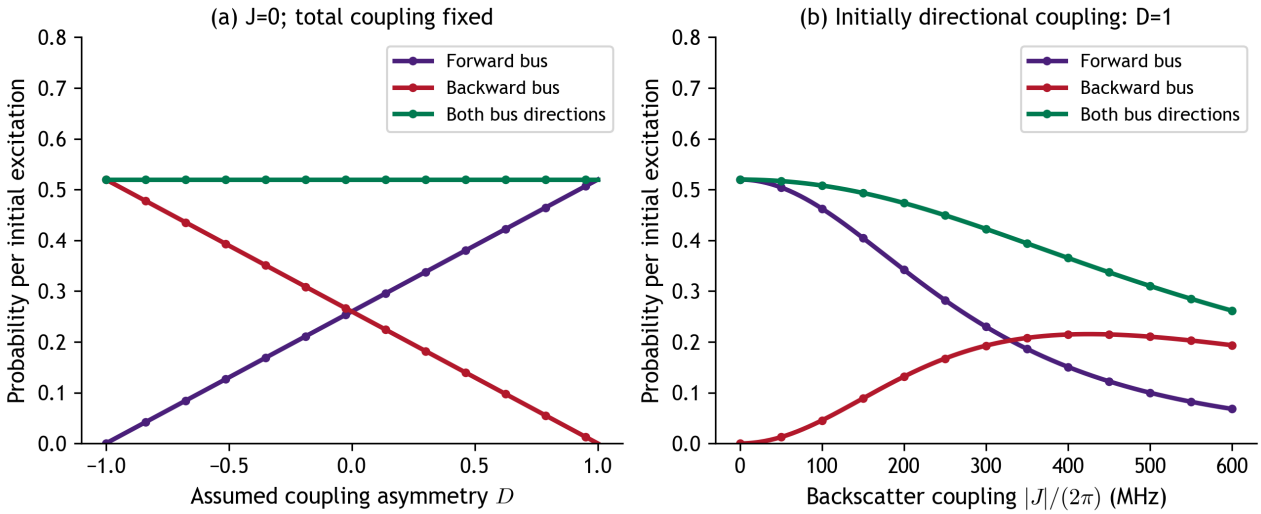
For an ideal reciprocal ring, the fields of the two time-reversed modes are complex conjugates. A real, linearly polarized transition dipole then couples equally strongly to both. Moving that molecule changes coupling phases, but position alone doesn't make it prefer one traveling direction.

A complex transition dipole, together with a locally elliptical field, can prefer a direction. The directional-coupling control lets you explore what that would do. Whether it applies to a particular molecule requires its transition polarization and the local vector fields.

$$P_R = \int_0^\infty \kappa_{\mathrm{ex},+} \left\langle a_+^\dagger a_+ \right\rangle dt, \quad P_L = \int_0^\infty \kappa_{\mathrm{ex},-} \left\langle a_-^\dagger a_- \right\rangle dt .$$

These are probabilities for the actual right and left outputs. If P_{bus} includes equal contributions from both directions and there is a grating at only one end, that grating receives half the bus light. If I have already calculated P_R, it already refers to the right output, so there is no additional

factor of one half. This matters when turning a ring-emission result into a collection budget. Molecular-ring experiments give a useful setting for thinking about the two paths. [R2]



Right and left emission from the two-direction ring model. The total bright coupling stays fixed while the directional preference or backscattering changes. Each output probability is integrated separately.

The bus carries a traveling optical field into and away from the coupling region. Input-output theory relates that field to the ring amplitude. With one consistent choice of port phase, the relation is

$$b_{\mathrm{out}}(t) = b_{\mathrm{in}}(t) - \sqrt{\kappa_{\mathrm{ex}}}\, a(t)\,.$$

For vacuum input, there is no incoming photon flux, and the emitted flux is $\kappa_{\mathrm{ex}}\langle a^{\dagger}a\rangle$. A laser-probe measurement is different. The light that stays in the bus interferes with the field leaking out of the ring, so both terms in b_{out} matter.

This is why a deep dip in through transmission doesn't directly tell me the collection probability of an initially excited molecule. The dip depends on interference with the incoming probe. The molecular-emission problem has no such probe field. [R4]

$$t(\Delta_L) = 1 - \frac{\kappa_{\mathrm{ex}}}{\kappa/2 - i\Delta_L}, \quad T = |t|^2, \quad t(0) = \frac{\kappa_i - \kappa_{\mathrm{ex}}}{\kappa_i + \kappa_{\mathrm{ex}}}\,.$$

This transmission formula describes an empty, single-bus ring with one traveling mode, using $\Delta_L = \omega_L - \omega_c$. At critical coupling, $\kappa_{\mathrm{ex}} = \kappa_i$, the through-port amplitude vanishes on resonance. The field coming from the ring cancels the directly transmitted field.

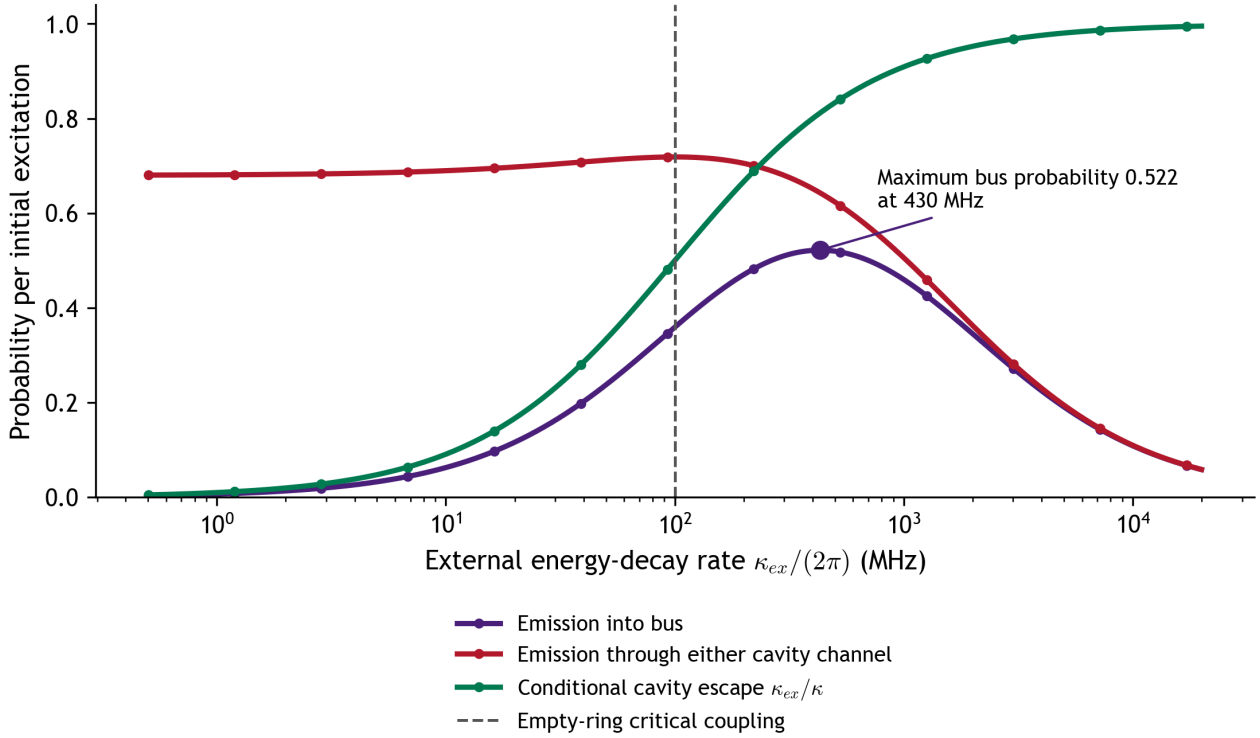
Critical coupling is therefore a condition for this probe experiment. It doesn't automatically give the best output from an excited molecule. For that, I care about both getting energy into the cavity and letting it escape into the bus.

Increasing κ_{ex} helps a cavity photon leave through the bus before intrinsic loss takes it away. But it also shortens the cavity lifetime and broadens the resonance. In the fast-cavity limit, that broadening reduces the on-resonance rate at which the molecule feeds the cavity. Those two effects compete, giving an optimum in the useful output probability:

$$P_{\mathrm{bus}} = \frac{\kappa_{\mathrm{ex}}\mathscr{R}}{\kappa\gamma_1 + (\kappa+\gamma_1)\mathscr{R}}, \quad \kappa = \kappa_i + \kappa_{\mathrm{ex}}\,.$$

The loading sweep changes κ_{ex} while holding g, the molecular rates and intrinsic loss fixed. That isolates the competition between feeding the ring and emptying it. Moving a real bus closer can also change the mode shape and intrinsic loss, so a geometry sweep may change more than one quantity.

The default example uses $g/2\pi = 100$ MHz, $\kappa_i/2\pi = 100$ MHz, $\kappa_{\mathrm{ex}}/2\pi = 500$ MHz, $\gamma_\phi/2\pi = 30$ MHz and a 5 ns background population lifetime. With one cavity mode and one bus output, the model gives a bus probability of about 0.520. You can change the inputs and watch where the optimum moves.



Bus output as the external coupling rate changes. The best molecular extraction occurs away from the empty-ring critical-coupling point. All other rates and the molecule-cavity coupling stay fixed in this sweep.

Once the photon is in the bus, propagation loss is a separate step. For a bus length L_b and power attenuation coefficient $\alpha_{P,b}$, the surviving fraction is $\eta_{\mathrm{prop}} = e^{-\alpha_{P,b}L_b}$. If the loss is quoted in dB per unit length, use $\eta_{\mathrm{prop}} = 10^{-\alpha_{\mathrm{dB}}L_b/10}$. The denominator is ten for a power ratio. A field-amplitude ratio uses twenty.

The grating has to turn a guided wave into an outgoing beam. The guided field accumulates phase along the chip faster than a freely propagating wave in the upper cladding can. The repeated teeth provide a way for light emitted at different positions to add in phase at a chosen angle.

Take light traveling along $+x$, with the wafer normal along $+z$. I'll measure the outgoing angle θ from $+z$ toward $+x$. For a periodic grating, the phase-matching condition is

$$\beta_B - \frac{2\pi m}{\Lambda} = k_0 n_t \sin\theta, \quad \Lambda = \frac{m\lambda_0}{n_B - n_t \sin\theta}.$$

Here Λ is the tooth-to-tooth pitch and m is the diffraction order. The term $2\pi m/\Lambda$ is the spatial phase contribution supplied by the periodic pattern. The grating changes the allowed in-plane wavevector while leaving the photon frequency unchanged.

The index $n_B = \mathrm{Re}\beta_B/k_0$ describes phase propagation through the grating, and n_t is the refractive index above it. In a periodic structure, the field pattern repeats from cell to cell with an accumulated phase -- this is the Bloch-like propagation described by β_B. The sign in the equation follows the directions chosen above.

For a weak grating, or one that changes slowly along its length, I can get a first pitch estimate by averaging the phase accumulated in the etched and unetched sections. If the unetched tooth occupies a fraction f of each period,

$$n_B \simeq f n_u + (1-f) n_{\mathrm{etched}}.$$

The two indices here are effective indices of guided sections. They aren't the bulk refractive indices of LN and air. This average leaves out strong interaction between repeated cells and any substantial reflection. I would use it to choose a starting pitch range, then calculate the periodic structure or full grating more carefully.

The toy starts at 595 nm with a 350 nm pitch, 18 periods and a chosen effective index of 2.05. In air, the first-order phase match is about $20.49°$, and the grating is 6.3 micrometres long. Changing pitch lets you see the outgoing angle move. The effective and group indices are inputs here; a later LN mode calculation would supply them.

For X-cut LN, I also need to say how the crystal sits in these coordinates. One useful choice puts crystal X_c along lab z, Y_c along x, and the optic axis Z_c along y. The relative permittivity tensor is then

$$\boldsymbol{\epsilon}_r = \mathrm{diag}(\epsilon_o, \epsilon_e, \epsilon_o).$$

In a strictly 2D model with no variation along y, TM has fields (E_x, E_z, H_y) and TE has (E_y, H_x, H_z). With the crystal assignment above, both electric-field components of strict 2D TM see the ordinary permittivity ϵ_o.

A finite-width guide can also have E_y, even when its dominant field is vertical and it is called TM-like. That component sees the extraordinary permittivity. This is one reason to look at the actual field components before carrying a 2D polarization label into a 3D design. The crystal tensor stays fixed in lab coordinates. [R9, R10]

Choosing the radiation angle is only half the grating design. I also want to choose how much light comes out at each position. That distribution sets the shape of the emitted beam. Let $A(x)$ be the guided field amplitude after separating out its propagation phase, and let $\alpha_A(x)$ describe the amplitude leaking away per unit length. Then

$$\frac{dA}{dx} = -\alpha_A A, \quad P = |A|^2, \quad \frac{dP}{dx} = -2\alpha_A P.$$

For constant leakage, the remaining guided power is $P(x) = P_0 e^{-2\alpha_A x}$. If all the power removed from the guide becomes radiation, the power emitted per unit length is $q(x) = 2\alpha_A P(x)$. The first teeth therefore emit more than the last ones, even when every tooth is identical. There is simply more guided power available near the entrance.

When adding the emitted fields, I use an amplitude proportional to $\sqrt{q}$. Power itself doesn't interfere. For discrete teeth, the same power accounting is

$$q_j = P_j(1 - e^{-2\alpha_{A,j}\Lambda}), \quad P_{j+1} = P_j - q_j.$$

The power emitted by tooth j is subtracted before the next tooth sees the field. The sum of emitted power and remaining guided power therefore stays equal to the input. In this toy, all removed power goes into the emitted array. For the real grating, I would also separate upward radiation, downward radiation, reflection, absorption and any other relevant outputs.

Suppose I'd like the emitted intensity to follow a chosen shape $f(x)$, with its integral over the grating set to one. I choose a total extracted fraction $\eta < 1$ and define $F(x) = \int_0^x f(s)\,ds$. The amount already emitted at position x is then $P_0\eta F(x)$. Subtracting it from the input gives the power still in the guide, and the leakage needed to produce the target shape:

$$P(x) = P_0[1 - \eta F(x)], \quad \alpha_A(x) = \frac{\eta f(x)}{2[1 - \eta F(x)]}.$$

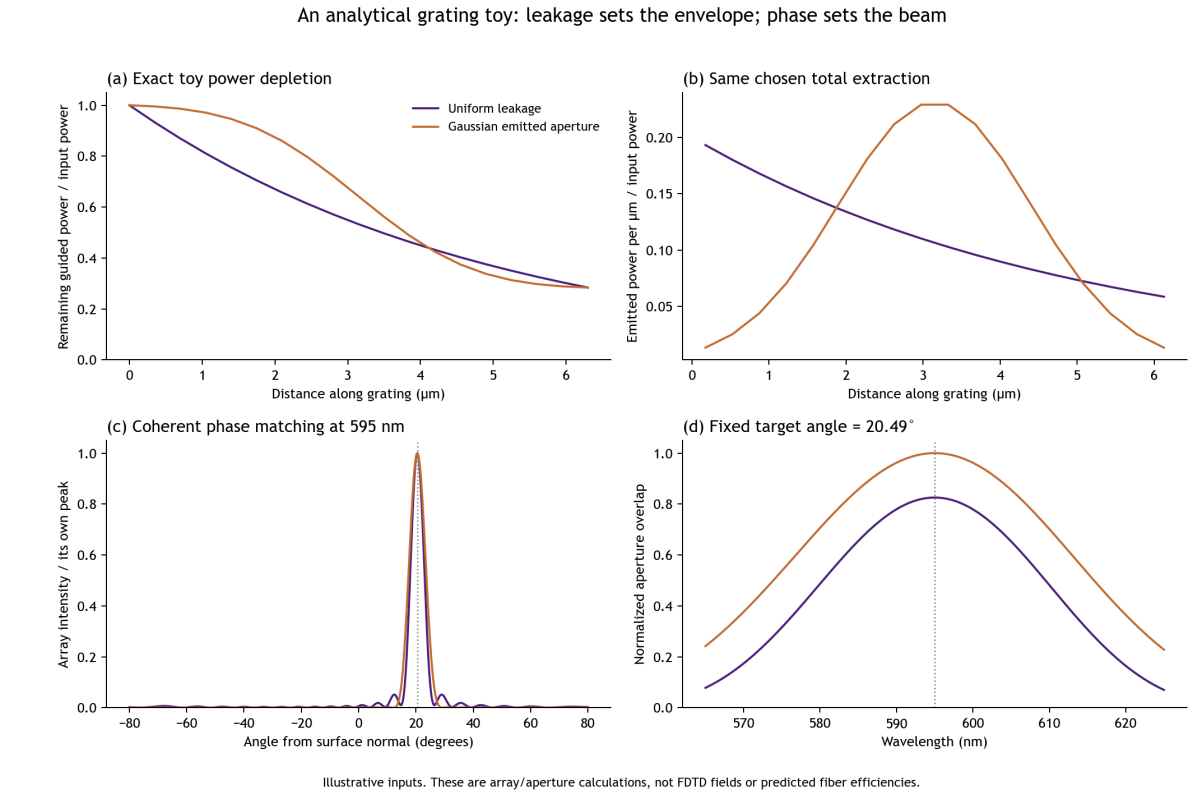
The denominator is the power still available at that position, apart from the common input factor. As the guide empties, a tooth has to extract a larger fraction of what remains to emit the same amount of light. This is why a Gaussian emitted envelope doesn't call for a Gaussian leakage profile.

Trying to extract absolutely everything over a finite length can make the required leakage near the end extremely large. Keeping $\eta < 1$ leaves some room. The discrete version, useful for the toy, is $\alpha_{A,j} = -\ln(1 - q_j/P_j)/(2\Lambda)$.

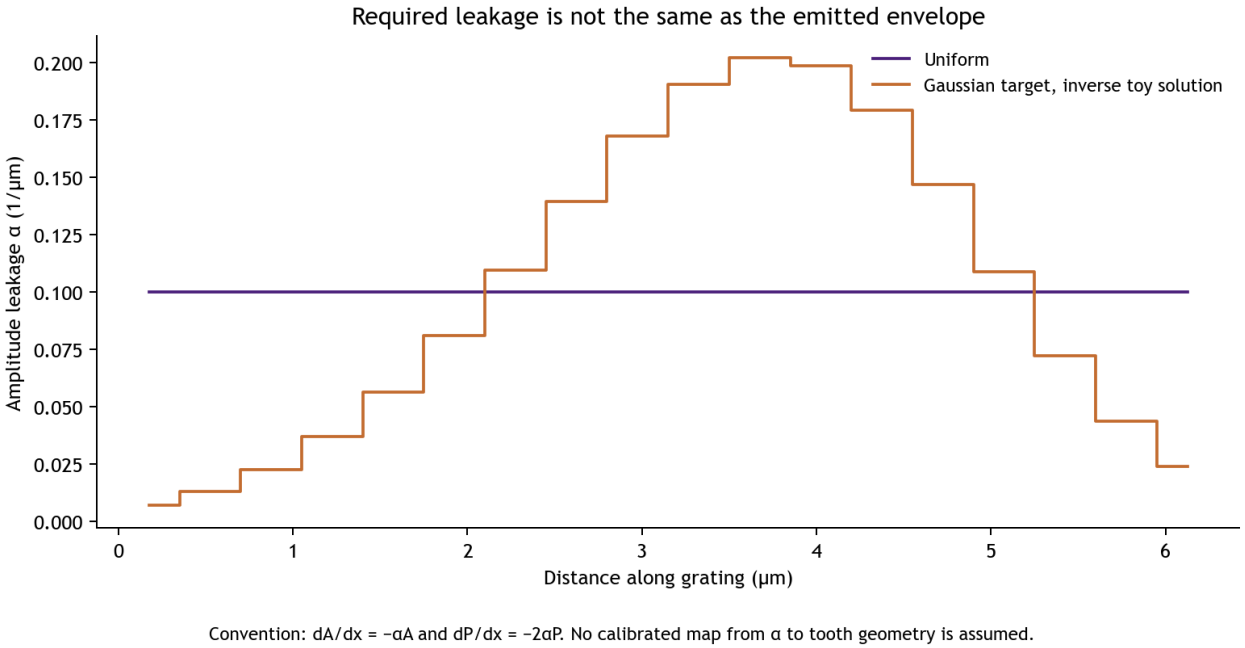
The toy compares constant leakage with a profile chosen to give Gaussian-shaped emission, keeping the total extracted fraction the same. Its waist w is defined for field amplitude, proportional to $e^{-(x-x_0)^2/w^2}$. The intensity therefore has twice that exponent.

At phase match, the designed aperture has unit normalized shape overlap with its chosen target because I constructed it to have that shape. That is a useful check on the construction. It doesn't tell me what fraction of a bus photon reaches a fiber, because upward directionality and the rest of the collection problem haven't been included in that shape comparison.

Varying the tooth geometry along a grating to shape the emission is called apodization. In a real device, changing tooth fraction or etch depth changes both leakage and propagation phase. To keep all the emitted light aimed in the same direction, I need both α_A and β_B as functions of geometry, then adjust the pitch or shape as needed. The toy shows the required optical profile; translating it into teeth is the next design problem. [R10]



The grating toy shows the power left in the guide, the emitted field envelope, the angular interference pattern and the normalized aperture overlap. Effective index and leakage are adjustable inputs.



Leakage needed to make the emitted intensity Gaussian. The guide has less power left toward the end, so the leakage profile differs from the desired emitted shape. Both examples extract the same chosen total power.

Light from all the teeth adds as a field, with a phase that depends on both the guided wave and the viewing angle. In a scalar model, the far-field amplitude is

$$E_\infty(\theta) \propto \int_0^L \sqrt{q_u(x)} \exp\left\{ i \int_0^x \left[\beta_B(s) - \frac{2\pi}{\Lambda(s)} \right] ds - i k_0 n_t x \sin\theta \right\} dx .$$

Here $q_u(x)$ is the upward-emitted power per unit length. The first phase term tracks the grating field, and the second accounts for the different paths from different emission positions to an observer at angle θ. At the matched angle, those contributions line up.

A flat emitted aperture gives a sinc-shaped angular amplitude. Its main intensity peak gets narrower as the aperture gets longer. The approximate full angular width at half maximum is

$$\Delta\theta_{\mathrm{FWHM}} \simeq \frac{0.886\lambda_0}{n_t L \cos\theta_0} .$$

This width uses a uniformly illuminated length L, with angles in radians. If most of the light leaks out near the entrance, the useful emitting length is shorter than the physical grating. Adding more teeth after the guide has emptied then does little to narrow the beam. Shaping the emission changes both the main lobe and the weaker side lobes.

The toy plots angular intensity relative to its own peak. It leaves out the polarization-dependent radiation of an individual tooth, transmission at material interfaces and diffraction across a finite guide width. It is useful for seeing interference and beam shaping, while an absolute collection calculation needs those other pieces too.

To couple into a fiber or a chosen free-space beam, the outgoing field has to match its phase and polarization as well as its intensity. Two beams can have the same intensity image and still couple poorly if their phases differ across the beam. I can compare the full electric and magnetic fields on a common plane in a homogeneous, lossless collection region. For an outgoing field F and target mode T, use

$$C(T,F) = \frac{1}{4} \int_S \left[\boldsymbol{E}_T^* \times \boldsymbol{H}_F + \boldsymbol{E}_F \times \boldsymbol{H}_T^* \right] \cdot \hat{\boldsymbol{n}} \, dA .$$

$$C(F,F) = P_F, \quad \eta_{\mathrm{GC}} = \frac{|C(T,F)|^2}{P_T P_{\mathrm{bus,in}}} .$$

The first equation above is a field overlap, with a power normalization. Putting the same field into both slots should give its power, $C(F,F) = P_F$. I use that as a quick check on the cross products and complex conjugates. For a valid decomposition into outgoing modes, the power in the target is $|C|^2/P_T$. Dividing by the incoming bus power gives η_{GC}.

Sometimes an overlap is instead reported as a fraction of upward power. In that case, I still have to multiply by the fraction of bus power that went upward. That factor belongs in the budget once. Keeping the input reference plane explicit avoids counting it twice.

Reciprocity gives another way to run the collection calculation. In a linear reciprocal structure, the coupling between a pair of reciprocal external modes is the same in either direction when the modes use compatible power normalization and boundaries. I can launch the guided mode and compare the outgoing field with the target beam, or send the reciprocal target beam onto the grating and measure the guided output.

The comparison has to use the same pair of modes. Reciprocity doesn't say that all upward radiation goes into the chosen fiber. It tells me how to reverse the particular coupling problem I want to measure. [R11, R12]

A 2D model assumes the structure extends indefinitely along its missing dimension. Absolute flux is then power per unit length in that direction. Power ratios can still tell me about that slab problem, including reflection and upward versus downward radiation. But the model cannot describe spreading across a finite width, the full fiber profile or the access taper in three dimensions. I'd use 2D to understand the grating and narrow the design choices, then include those missing dimensions in the collection model.

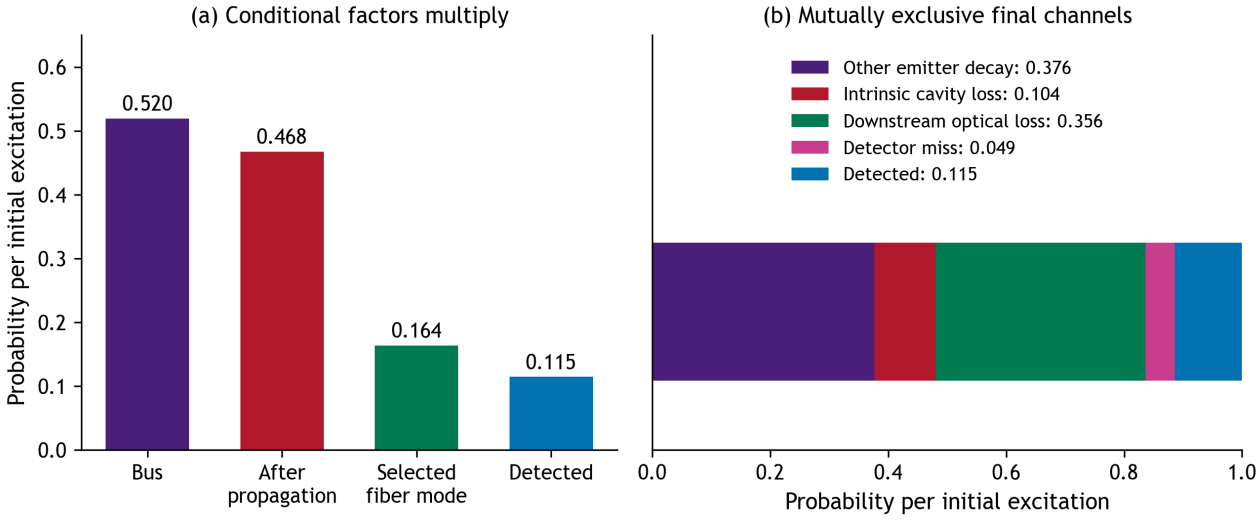
I can now put the path together. Start with one excited molecule, let the selected bus output receive its share of the emission, and follow the photon through the rest of the optics. If light travels downstream without appreciable feedback, the detection probability is

$$P_{\mathrm{det}} = \left[\int_0^\infty \kappa_R p_c(t)\, dt\right] \eta_{\mathrm{prop}} \eta_{\mathrm{GC}} \eta_{\mathrm{opt}} \eta_{\mathrm{det}}\,.$$

The bracket is the probability that the initial excitation produces a photon in the chosen bus channel. Each factor after it is conditional on having reached the previous step. η_{prop} accounts for bus propagation, η_{GC} for coupling from the incoming bus mode into the target optical mode, η_{opt} for subsequent optical transmission, and η_{det} for detection.

This starts after successful preparation. If the molecule is prepared in the excited state in a fraction p_{prep} of trials, the per-trial detection probability in this preparation model is $p_{\mathrm{prep}} P_{\mathrm{det}}$.

For a ring with two directions, I use the integral for the output I actually collect, or add the outputs if both are routed into the experiment. A grating at the right end doesn't collect the left-going photon just because both photons came from the same ring. The budget plot makes this choice explicit.



Collection starting from one excited molecule, using one cavity mode and one bus output. For an equally coupled reciprocal ring with only one collected arm, the detected probability is half this single-output case. The grating efficiency here is an assumed input to the budget, separate from the normalized aperture overlap in the grating toy.

Multiplying by single efficiency numbers assumes those efficiencies are nearly constant across the photon spectrum. If they vary appreciably, I weight them by the spectrum instead. Let $S_R(\omega)$ be the spectral probability density of the selected bus photon, so that its frequency integral gives the selected bus probability. Then

$$P_{\mathrm{det}} = \int S_R(\omega)\eta_{\mathrm{prop}}(\omega)\eta_{\mathrm{GC}}(\omega)\eta_{\mathrm{opt}}(\omega)\eta_{\mathrm{det}}(\omega)\,d\omega\,.$$

Finding S_R requires the emitted two-time field correlation discussed earlier. The population trace tells me how much light leaves over time, but doesn't contain all the phase information needed for the spectrum.

There is also a physical limit to treating the rest of the chip as independent efficiency factors. A grating reflection can send a coherent field back toward the ring. If that return is appreciable, it can change the ring dynamics and introduce a delay. I would then include that path in the dynamical model instead of attaching it as a final transmission factor.

When checking a classical grating simulation, I want every part of the input power accounted for exactly once. For example,

$$1 = R + T_{\mathrm{guided}} + \eta_{\uparrow} + \eta_{\downarrow} + \eta_{\mathrm{abs}} + \eta_{\mathrm{other}}\,.$$

Each term has to refer to a separate destination. Suppose a downward monitor above a silicon substrate measures power entering the substrate region, and I count that as the downward output. If the same light is absorbed later in the silicon, adding that absorption again would count it twice.

At 595 nm, I also need silicon optical data appropriate to the visible range. A lossless material model borrowed from a telecom example could miss an important channel. These material choices affect the power budget before any fiber-overlap calculation.

With that picture in place, each Tidy3D calculation has a question to answer. A waveguide mode solve gives the phase propagation, group index and vector field. A ring–bus coupler calculation gives the fraction transferred per pass, which converts to κ_{ex}. A ring calculation gives its resonant frequency, energy decay and normalized mode field. A dipole run shows how the local environment responds at the molecular site and how the emitted power divides among channels. A grating run connects an incoming guide mode to outgoing optical modes.

I don't need all those calculations at once. The toys help identify which quantity is driving the result, so the next electromagnetic run has a clear purpose.

Tidy3D uses micrometres for geometry, seconds for time and hertz for frequency. A wavelength of 595 nm is therefore 0.595 in its length units, and the corresponding frequency is td.C_0 / 0.595. I'd put the units in parameter names and plot labels from the start. Material dispersion and frequency arrays also need to follow the library's conventions. [R6]

A simulation needs the region being modeled, its materials and shapes, a source, a grid, boundary conditions, monitors and a run time. A ModeSource launches a chosen guided mode. A PointDipole drives a local polarization current. FieldMonitor records fields, while FluxMonitor records signed power crossing a surface. A ModeMonitor separates the field into specified guided modes and propagation directions.

That last distinction is useful at an output plane. Some light crossing the plane may be radiation or a mode I don't want. If the next component accepts one guided mode, its power is the relevant output, rather than all the flux through the plane.

I'd draw the coordinates and crystal axes before selecting a polarization filter. Then solve the cross-section and look at the actual electric-field components and where the energy sits. Mode index zero is just a place in the solver's sorting order; it doesn't always mean the same physical TM mode.

As geometry changes, modes can approach and mix near an avoided crossing. Following the field overlap, location and polarization helps track the branch of interest rather than accidentally jumping to another mode with a similar index.

For a dipole run, specify the molecular position, orientation and host, and keep the same source convention in the reference calculation. Separate radiation, absorption where relevant, and

guided output modes. If the quantum model keeps a narrow cavity resonance explicitly, separate that contribution from the background decay rate before assigning γ_1.

The division of work is useful here. Tidy3D supplies the electromagnetic response of the structure. The molecule–cavity model uses that information to evolve the excited state and calculate emission probabilities or photon statistics.

For a ring, I'd look at the decay after the excitation source has stopped. An isolated field amplitude proportional to $e^{-\kappa t/2}$ has intensity proportional to $e^{-\kappa t}$. A clean exponential can give the energy decay rate and Q.

If several modes are beating together, or the source is still driving a transient, one exponential fit can be misleading. When comparing grids or boundaries, I want to follow the same resonance and normalize the fields in the same way.

For the grating, I'd launch the input guide mode and place mode monitors before and after it to measure reflection and guided transmission. Upward and downward flux surfaces track where the radiation goes. A field plane in the collection region supplies the data for the target-mode overlap.

The target beam needs its waist, tilt, phase front and polarization specified. That gives the simulation a definite collection question: how much input bus power reaches this particular optical mode?

The outer boundary usually uses a PML, or perfectly matched layer. It is an artificial absorbing region intended to let outgoing waves leave without coming back into the device. I would change the buffer distance between device and PML separately from the grid spacing, because boundary error and resolution error are different effects.

Small etched features and large changes in refractive index can need a finer local grid and suitable treatment of interfaces that fall between grid points. The run time has to cover the slowest dynamics of interest. A high-Q mode may keep ringing long after the source pulse ends, so source shutoff alone isn't a reason to stop the calculation.

The official grating examples are useful for seeing how these objects fit together and how a 2D study can lead into a 3D design. Their wavelengths, materials and beam targets belong to those examples, though. I'd choose the visible-wavelength LN material inputs and physical coordinates for this design separately. For now, the supplied examples are local analytical models, so you can explore them without launching an FDTD job. [R9, R10, R12]

To get started, create a personal account through the Tidy3D signup page. Use your university identity when applying for educational access, then follow Apply on Flexcompute's Educational

Licenses page to the account Plan tab. The Education Free should give 10 FlexCredits each month for free. If your educational application is approved, the account page is the place to check the allowance and expiry that apply to you. [R13]

For Python access, install Tidy3D in a clean environment, get your API key from the account, and enter it at the tidy3d configure prompt. The current installation instructions walk through checking the connection. The key belongs in your private configuration, rather than in a shared script or this guide. [R6]

Building a model locally and running a cloud solve are separate steps. Before submitting a run, look at the estimated maximum credit charge and your available balance. Afterward, check the actual charge. The account page also gives the plan details, including renewal and any other requirements. [R7]

The three toys in this package use NumPy, SciPy and Matplotlib on your own computer. They don't use Tidy3D credits. Their README files give the setup and run commands. gc_toy.py explores the grating, bus_coupler_toy.py explores spatial transfer between the ring guide and bus, and qed_toy.py follows the molecular excitation through the ring outputs. The JSON files hold the inputs, and each run saves parameters, CSV data and plots.

For the grating, I'd first choose a collection angle and enter it as target_angle_deg. Then change the pitch and watch what happens at that fixed angle. If target_angle_deg stays null, the toy moves the target to the new phase-matched angle on every run. That asks how well each grating works with its own best direction, rather than what a fixed collection optic receives.

Try period count next, then leakage. More periods change the available emitting length, while stronger leakage makes the entrance contribute more heavily. The duty control changes leakage through a chosen first-harmonic law. It doesn't yet calculate how the actual tooth geometry changes LN effective index or reflection.

For the quantum model, I'd start with the dipole-angle and height plots. Apply the field projection you want to the bright coupling in the JSON inputs, then explore detuning and pure dephasing. After that, change the external loading while keeping intrinsic loss and coupling fixed. Compare the population traces with the eventual output probabilities, and then compare one selected output with the two-direction ring.

The collection budget uses 0.35 as an adjustable grating-efficiency assumption in one example. Changing the separate grating toy doesn't automatically update that number. Its aperture overlap describes shape matching; the budget needs the full bus-to-target efficiency.

The aim is to get a feel for how the pieces fit together, with enough math that you can check the reasoning and change the assumptions. You can regenerate the plots, edit the manuscript, or replace a toy input with a quantity from a later field calculation. I'd keep asking what each number describes -- a field amplitude, a probability, a rate, or a fraction of power -- because that makes the connections between the models much clearer.

The first model starts with one excitation and no laser afterward, so it can emit at most one photon. That is built into the initial condition. It doesn't tell me the chance of producing two photons during a real excitation pulse.

To study that, I add the laser drive and allow enough molecular and cavity states to hold the excitations that can occur. For a drive on the molecular transition, the added Hamiltonian is

$$\frac{H_d(t)}{\hbar} = \frac{1}{2}[\Omega(t)\sigma_+ + \Omega^*(t)\sigma_-] .$$

The function $\Omega(t)$ describes the drive strength and phase over the pulse. Suppose the molecule emits before the pulse is over. It is back in the ground state while the laser is still present, so it can be excited again and emit another photon. Being a single molecule doesn't prevent two emissions at different times within the same pulse cycle.

Faster emission can therefore affect preparation and re-excitation, as well as the repetition rate. A static Purcell factor can't answer the whole finite-pulse question. [R5]

One way to calculate the photon-number distribution is to keep a separate density matrix ρ_n for histories with n emissions into the channel being counted. These matrices are not individually normalized; their traces are the probabilities P_n. Let $\mathscr{L}$ be the full trace-preserving density-matrix generator, and let $\mathscr{J}\rho = L_{\mathrm{count}}\rho L_{\mathrm{count}}^{\dagger}$ be the counted jump. Then

$$\dot{\rho}_n = (\mathscr{L} - \mathscr{J})\rho_n + \mathscr{J}\rho_{n-1}, \quad P_n = \mathrm{Tr}\rho_n .$$

The counted jump moves probability from the $n-1$ group to the n group. Losses that aren't counted still affect the system dynamics through $\mathscr{L}$. After the pulse and subsequent decay, the number probabilities give the mean count and the pulsed second-order correlation:

$$\mu = \sum_n n P_n, \quad g^{(2)}_{\mathrm{pulse}} = \frac{\sum_n n(n-1)P_n}{\mu^2} .$$

The factor $n(n-1)$ counts ordered photon pairs. It is zero for zero or one photon, so the numerator is sensitive to events with more than one counted photon. This pulse-integrated quantity differs from the instantaneous $g^{(2)}(0)$. To compare with an experiment, I would use its collection channel, detection model, background and time window. The classical dipole calculation

supplies the passive optical response needed by such a model, but doesn't supply the photon-number distribution.

More molecular states can be added when they affect the experiment. If a vibrational state relaxes much faster than the dynamics of interest, it may be possible to remove it explicitly and include its effect through rates. A long-lived triplet state is different: a molecule can spend several cycles there without fluorescing, giving memory and blinking across trials.

The transition frequency can also drift, a process called spectral diffusion. If that drift is slow compared with one emission event, I can calculate the event at a fixed detuning and average over a distribution of detunings afterward. If the frequency changes during the event, the model needs the time-dependent fluctuations. The right choice depends on the molecule and host, rather than on a universal correction factor.

[R1] Wang et al. Coherent coupling of a single molecule to a scanning Fabry Perot microcavity. https://arxiv.org/pdf/1612.05007

[R2] Rattenbacher et al. Coherent coupling of single molecules to on chip ring resonators. https://arxiv.org/html/1902.05257

[R3] Cui and Raymer. Emission spectra and quantum efficiency of single photon sources in the cavity QED strong coupling regime. https://arxiv.org/html/quant-ph/0601108

[R4] Gardiner and Collett. Input and output in damped quantum systems. https://doi.org/10.1103/PhysRevA.31.3761

[R5] Fischer et al. Pulsed Rabi oscillations in quantum two level systems beyond the area theorem. https://arxiv.org/html/1708.05444

[R6] Tidy3D installation and API configuration. https://docs.flexcompute.com/projects/tidy3d/en/latest/install.html

[R7] Tidy3D frequently asked questions and source normalization. https://www.flexcompute.com/tidy3d/learning-center/faq/

[R8] Sauvan et al. Theory of the spontaneous optical emission of nanosize photonic and plasmon resonators. https://arxiv.org/pdf/1304.8110

[R9] Tidy3D uniform linear grating coupler. https://docs.flexcompute.com/projects/tidy3d/en/latest/notebooks/GratingCoupler.html

[R10] Tidy3D focusing apodized grating coupler. https://docs.flexcompute.com/projects/tidy3d/en/latest/notebooks/FocusedApodGC.html

[R11] Tidy3D mode overlap and vector field projection. https://docs.flexcompute.com/projects/tidy3d/en/latest/notebooks/ModeOverlap.html

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[R13] Flexcompute educational licenses. https://flexcompute.com/educational-licenses

[R14] Grange et al. Cavity funneled generation of indistinguishable single photons from strongly dissipative quantum emitters. https://arxiv.org/html/1501.00931